

Sampling Rate and Bit Depth

Audio Features and Convolutional Neural Networks

Deployment to Embedded Systems

Project and Review



Reading: Project - Sound Classification
2h



Quiz: Audio Classification
10 questions



Discussion Prompt: Share Your Audio Classification Project!
15 min



Course End Survey
5 min



Video: Conclusion
2 min



Project - Sound Classification

Introduction

In this module, we went through the steps of creating a speech recognition system that can be used to create a device that recognizes and classifies other sounds.

In this project, we will start with a simple, pre-made dataset, add our own sounds, and encourage you to find sounds made with something other than your voice.

Please note that at this time, we do not have a way to grade your project for the course. However, we strongly encourage you to go through these steps to learn as much as possible, learning as well as using the Edge Impulse tool.

Required Hardware

For collecting sound data, you should have access to a recording device.

For deploying, you can use either your smartphone or the [Arduino Nano](#).

The smartphone will provide a simple demo only whereas the Arduino board can handle more complex audio events. As a result, we recommend using the Arduino, if you have access to one.

Collect Data

We will start with a pre-made dataset from Edge Impulse that includes a variety of sounds running. Initially, this dataset was intended to be used as a standalone dataset. You are welcome to use just that data, augment it with your own sounds, or create your own dataset.

To start, download the faucet dataset from [this link](#). Unzip it somewhere on your computer.

If you are going to use your own sounds, I recommend collecting at least 10 recordings (at least one 50 second recording). Make sure the sound source is in different locations relative to your recording device. This will help create a more robust model that can differentiate between different sounds.

For example, I recorded a fan running at different speeds. I moved the microphone to record the sound in front of, behind, and above the fan, as the sound changes as the fan speed changes.

